

Muhammad Shaheer Luqman

AI researcher working in reinforcement learning, computer vision, and robot manipulation. Published across arXiv (2x) and IEEE, ICPC Gold Medalist, and summa cum laude graduate (3.90/4.00).

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Research & Publications

Enhancing Human-Likeness in Reinforcement Learning Agents via Hierarchical Macro Action Quantization 2026
[arXiv:2605.30928](https://arxiv.org/abs/2605.30928)

Under Review

A human-like RL framework (HiMAQ) that encodes human demonstrations into a two-level hierarchy of macro actions for dexterous humanoid-hand control. On D4RL it improves human-likeness over the non-hierarchical baseline while matching or exceeding success rates across IQL, SAC, and RLPD. [View on arXiv](#)

Unsupervised Skeleton-Based Action Segmentation via Hierarchical Spatiotemporal Vector Quantization 2026
[arXiv:2604.15196](https://arxiv.org/abs/2604.15196)

Under Review

A hierarchical spatiotemporal vector-quantization framework for unsupervised skeleton-based action segmentation, achieving state-of-the-art on the HuGaDB, LARa, and BABEL benchmarks. [View on arXiv](#)

Ensuring Child Safety: An IoT-Based Surveillance System for Remote Monitoring and Detection of Anomalous Behavior 2024
[ICETST 2024 • IEEE Xplore](#)

An IoT surveillance system pairing hardware sensors with AI-driven anomaly detection for real-time child-safety alerts. [View on IEEE Xplore](#)

Research Interests: Robot Manipulation, Computer Vision, Reinforcement Learning, Vision-Language Models, IoT + AI Systems

Research Experience

Applied Research Engineer — Retrocausal Jun 2025 – Present
US-based • Remote

- Research on robot manipulation and Vision-Language-Action systems, unsupervised action segmentation, and temporal long-video understanding — **two papers submitted** (see above).
- Built and shipped end-to-end AI systems for industrial clients (incl. **BMW**), from model training to GPU deployment on **AWS** and **RunPod**.

Reinforcement Learning Robotics VLMs Computer Vision PyTorch

Software Engineer Intern — Retrocausal Mar 2024 – Jun 2024
Internship • US-based • Remote

- Built a video activity-recognition pipeline with **body pose estimation** and **vision-language models** for automated industrial workflow analysis.
- Developed **LLM prompt-engineering** systems for analysis of industrial process data.

Computer Vision VLMs LLMs MediaPipe Python

Education

BS Computer Science — FAST NUCES Sep 2021 – Jun 2025
CGPA: **3.90 / 4.00** — **Summa Cum Laude Honors**

A Levels — Generation's School Sep 2019 – Jun 2021
Grades: A*, 2A

IELTS Academic — English Proficiency Jul 2026
Overall Band: **7.5 / 9** • Reading **9.0** • Listening **8.5**

Skills

Research Areas Reinforcement Learning (IQL, SAC, RLPD); Robot Manipulation; Computer Vision; Vision-Language Models; Action Segmentation.

ML Frameworks PyTorch, Hugging Face, Scikit-learn; YOLO, MediaPipe; Stable Diffusion; BERT, Whisper.

Engineering Python, C / C++; Django, FastAPI, Flask; AWS, RunPod; Docker, Git, Linux.

Selected Project

PixelForge — Generative AI Marketing Platform

Final Year Project ([GitHub](#))

Full-stack generative pipeline combining **BLIP** captioning, **Gemini** text generation, and diffusion-based image and video synthesis for product-aware marketing content.

More on my [portfolio](#) and [GitHub](#).

Honors & Awards

- **Summa Cum Laude** — 3.90 / 4.00 CGPA, FAST NUCES (2025)
- **ICPC Gold Medalist** — ICPC National Competitive Programming (Nov 2024)
- **ICPC Bronze Medalist** — ICPC National Competitive Programming (Feb 2024)
- **4× Rector's List** (4.0 GPA) • **4× Dean's List** (3.5+ GPA) — FAST NUCES